

# Exam practice problems

I suggest you do some of these problems, as needed, to practice for the various exams you'll have to take in Applied Math. I'm happy to discuss them; I won't publish solutions. I'll put similar questions on the midterm, the final exam, and the qualifying exam.

## 1 week 1

The topics we covered in lecture and associated homework were:

- Ordinary generating functions: polynomial, rational, algebraic, d-finite, the types of sequences they enumerate, and the combinatorial meaning of algebraic operations (sum, product, derivative, integral, multiplication by  $1/(1-x)$  and  $(1-x)$ )
- permutations, the inversion generating function
- probability generating functions, expected value
- experimental math / "guess a recurrence relation/DE"
- Catalan objects
- Sicherman dice

Here are some questions - I think many of these would be a little long for exam questions:

1. A finite sequence is *palindromic* if it is the same written forwards and backwards. A polynomial is palindromic if its sequence of coefficients is palindromic. Prove that a degree  $d$  polynomial  $f(x)$  is palindromic if  $f(x) = x^d f(1/x)$ . Then prove that the inversion generating function for permutations of  $n$  is palindromic. (is there a bijective proof? I am a combinatorialist so I am contractually obligated to say that. But I don't care what the answer is.)
2. Prove that the generating function for inversions of a permutation in  $\mathbb{S}_n$  is

$$\prod_{i=1}^n \frac{1 - q^i}{1 - q}$$

3. Find a sequence of positive integers  $a_n$  whose ordinary generating function is rational, and for which  $a_n \sim C \left( \frac{5+\sqrt{13}}{2} \right)^n$ .
4. Suppose that  $a_0 = 0$ ,  $a_1 = 2$ , and that  $a_n = 4(a_{n-1} - a_{n-2})$  for  $n \geq 2$ . Find a closed form for  $a_n$ .

5. Suppose that  $a_n$  is a sequence such that  $a_n = -a_{n-1} - a_{n-2}$  for  $n \geq 2$ . Without finding an explicit closed form for  $a_n$ , explain why the sequence  $a_n$  repeats with period at most 6.
6. Suppose that  $a_n$  is a sequence of rational numbers satisfying

$$a_0 = 1, a_1 = 0, \text{ and } (n^2 - n)a_n + 9a_{n-2} = 0 \text{ for all } n \geq 2.$$

Find a closed form for the ordinary generating function  $G(x)$ . Then, find a closed form for the sequence of coefficients  $a_n$ . Hint: what class of nice generating functions does  $G$  fall into, based on the form of the recurrence?

7. Everybody knows that  $\frac{d^n}{dx^n}(1-x)^{-1} = n!(1-x)^{-n-1}$ , but what sequence of numbers has OGF  $(1-x)^{-n-1}$ ?
8. Suppose that  $f(x)$  is a polynomial of degree  $d$ , and that  $a_n$  is the sequence of values of  $f$ :  $a_n = f(n)$ . Describe the generating function of the sequence  $\{a_n\}_{n \geq 0}$  as succinctly as you can. What factors can appear in the denominator?
9. Construct a pair of 4-sided sicherman dice (two four-sided dice with unusual face numberings which, when rolled together, give the same probability distribution as two ordinary 4-sided dice).
10. A Dyck path of length  $2n$  is a path from  $(0,0)$  to  $(n,n)$  in the integer lattice made of length one steps up or to the right, such that the path never crosses strictly below the line  $y=x$ . Show that the generating function for Dyck paths  $G(x)$  satisfies  $G = 1 + xG^2$ , and hence that the number of Dyck paths of length  $2n$  is the  $n$ th Catalan number.
11. let  $C$  be the speed limit on a Canadian road in km/h, and let  $A$  be the speed limit on a similar road in the USA in miles/h. Why is  $(C,A)$  often a pair of adjacent Fibonacci numbers (with an extra zero)? For instance:
  - the speed limit in a school zone is 20 mph in the usa, 30 km/h in Canada.
  - For city streets, it's 30 mph in USA, 50 mph in Canada.
  - For old, low quality highways, it's 50 mph in USA, 80 mph in Canada.

## 2 week 2

The topics I covered:

- Given finitely many objects whose size is marked by  $x$  in the generating function  $G(x)$ , you can find the average size by taking

$$x \frac{d}{dx} \log(G(x)) \Big|_{x=1}$$

and a similar statement holds for probability generating functions.

- Integer partitions.
- analysis of algorithms (we did bubblesort; you did mergesort): best, worst, average case. Big O notation.
- Optimizing algorithms.
- Iteration for combinatorial objects; generator expressions; recursion.
- randomly permuting a list (fisher-yates, naively).

Questions for Week 2:

1. Write down how to find the mean of a random variable, given its probability generating function.
2. Write down how to find the *variance* (i.e. second moment) of a random variable, given its probability generating function. First do the case where the mean is zero.
3. Describe pseudocode for listing every involution on the set  $1, \dots, n$  (an involution is a permutation  $\pi$  such that  $\pi^2 = id$ .)
4. How many involutions on  $1, \dots, n$  are there?
5. Describe in pseudocode, an iterator for the  $n$ -fold cartesian product of an interval  $[0, \dots, k]^n$ .
6. Calculate the expected number of inversions in a randomly chosen permutation in  $\mathbb{S}_n$ , using the fact that the generating function for inversions of a permutation in  $\mathbb{S}_n$  is

$$G(x) = \prod_{i=1}^n \frac{1-x^i}{1-x}$$

7. The Poisson distribution is a probability distribution on  $\mathbb{Z}$  with parameter  $\lambda$ . It's characterized by the following: if  $X$  is a poisson-distributed random variable, then

$$\text{prob}(X = k) = \frac{1}{k!} \lambda^k e^{-\lambda}.$$

Let  $G(x)$  be the probability generating function for the Poisson distribution. Find a simple closed form for  $G(x)$ ; then find the expected value of the Poisson distribution.

8. Let  $L$  be a sorted list of numbers of length  $2^k - 1$  and let  $a$  be your favorite number. We want to determine whether  $a \in L$ . Here's an algorithm for that, called "binary search". Let  $b$  be the middle element of  $L$  (there is one). If  $a < b$ , then binary search for  $a$  in the first half of  $L$ . If  $a = b$ , then  $a$  is in  $L$ . If  $a > b$ , then binary search for  $a$  in the second half of  $L$ . Analyze the running time of binary search.

9. If you need to search a list repeatedly for different numbers, at what point (roughly) does it make sense to sort the list first and use binary search?
10. Let  $a, b$  be positive constants. Show that  $a^n$  is  $O(b^n)$  if and only if  $a \leq b$ .
11. Let  $W = \{0, 1\}^n$  be the set of binary words on  $n$  letters. Describe an algorithm for iterating over the words in  $W$ , producing words  $w_1, w_2, \dots$  such that  $w_i$  differs from  $w_{i-1}$  in exactly 1 digit, as do  $w_1$  and  $w_{2^n}$ . (This is called a *binary gray code of length  $n$* ; it's equivalent to finding a Hamiltonian cycle on the  $n$ -cube graph).
12. Suppose  $a_n$  is the average number of parts of a partition of  $n$ . Write down the generating function for  $\{a_n\}$ . Start by writing down a two-variable generating function which keeps track of both  $n$  and the number of parts, and simplify it as much as you can. I don't think there's a closed form, but I was able to write down an expression with just one sum in it.
13. Prove, using generating functions, that the number of partitions of  $n$  into distinct parts is equal to the number of partitions of  $n$  into odd parts (this is a famous problem due to Euler).